Review: Similar figures

1 Identify the unit used to express the scale factor.

2 If a scale factor of $\frac{2}{3}$ is applied to a shape, will the new shape be larger or smaller than the original?

3 If an image is enlarged by 250%, find the scale factor as a decimal value.

4 Determine whether the following are true when you enlarge or reduce a shape.

a Multiply each side by the same factor.

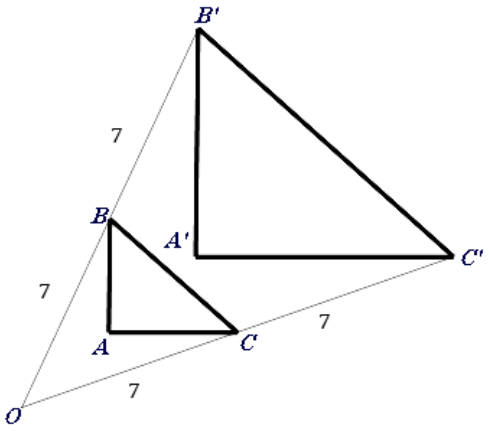
b Vary the side lengths and the angles.

c Multiply each angle by the same factor.

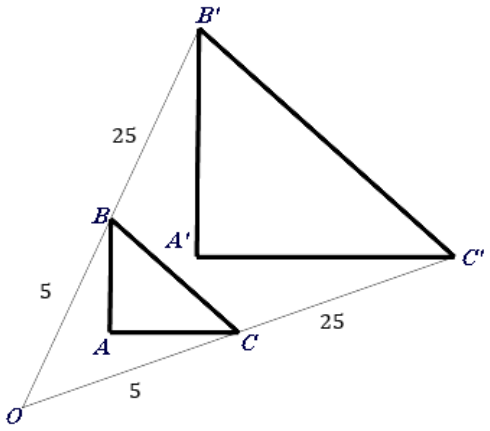
d Vary the angles but not the side lengths.

5 Given that $\triangle ABC$ has been enlarged to $\triangle A'B'C'$, find the scale factor of the following diagrams.

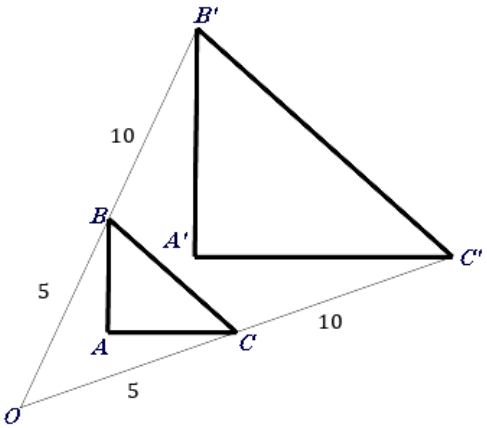
a



b

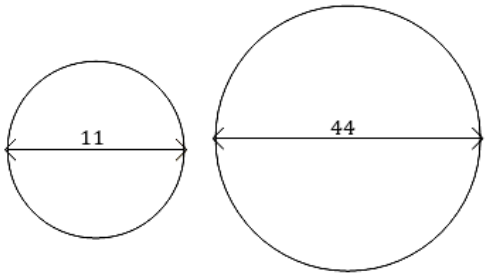


6 $\triangle A'B'C'$ has been reduced to form a smaller $\triangle ABC$.
Identify the reduction factor.

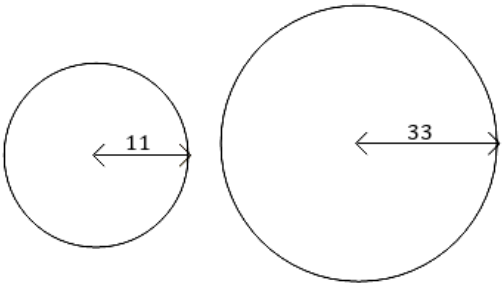


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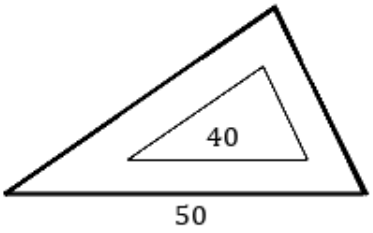
7 Determine the enlargement factor for the two circles in the diagram.



8 Determine the reduction factor for the two circles in the diagram.



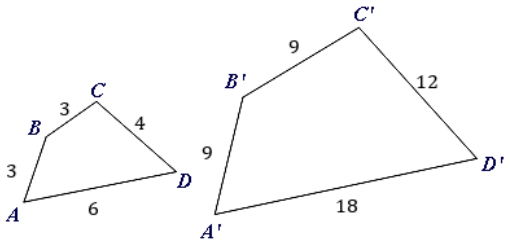
9 The scale factor of the following similar figures is $40 : 50 = 4 : 5$.
Identify the reduction and enlargement factors.



10 Glass in the shape of a circle has a radius of $12\ \text{cm}$. When it is blown, the radius becomes $48\ \text{cm}$.
Find the enlargement factor.

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11 Consider the following quadrilaterals.

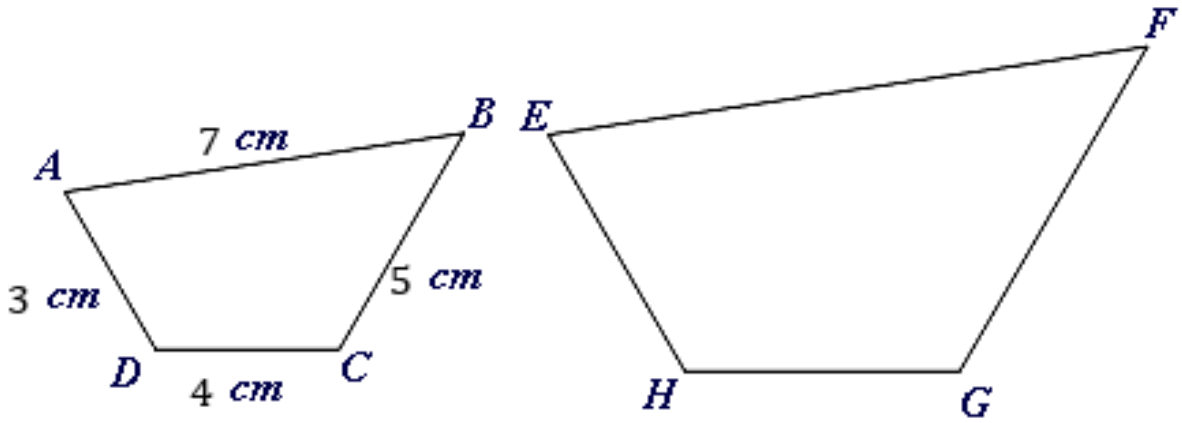


a Determine whether quadrilateral $A'B'C'D'$ is an enlargement of quadrilateral $ABCD$.

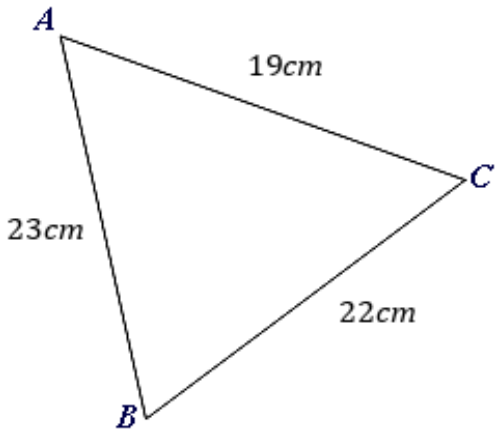
b Find the enlargement factor.

12 Mae wants to insert a picture into a document. She enlarges it by a factor of 12 but it becomes too blurry, so she reduces the resulting picture by a scale factor of 4. Find the overall scale factor going from the original to the final size.

13 In the diagram below, the scale factor used to enlarge the smaller quadrilateral is 4.5. Find FG.



14 The dimensions of the triangle shown are enlarged using a scale factor of 3.1.



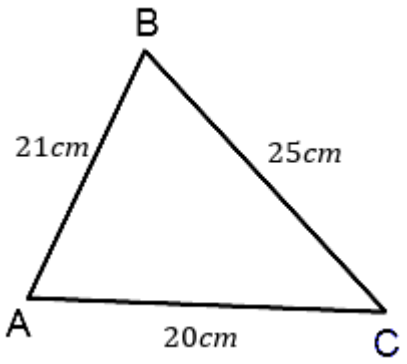
a Find the new length of AB.

b Find the new length of BC.

c Find the new length of CA.

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15 The dimensions of the triangle shown are reduced using a scale factor of 0.92.

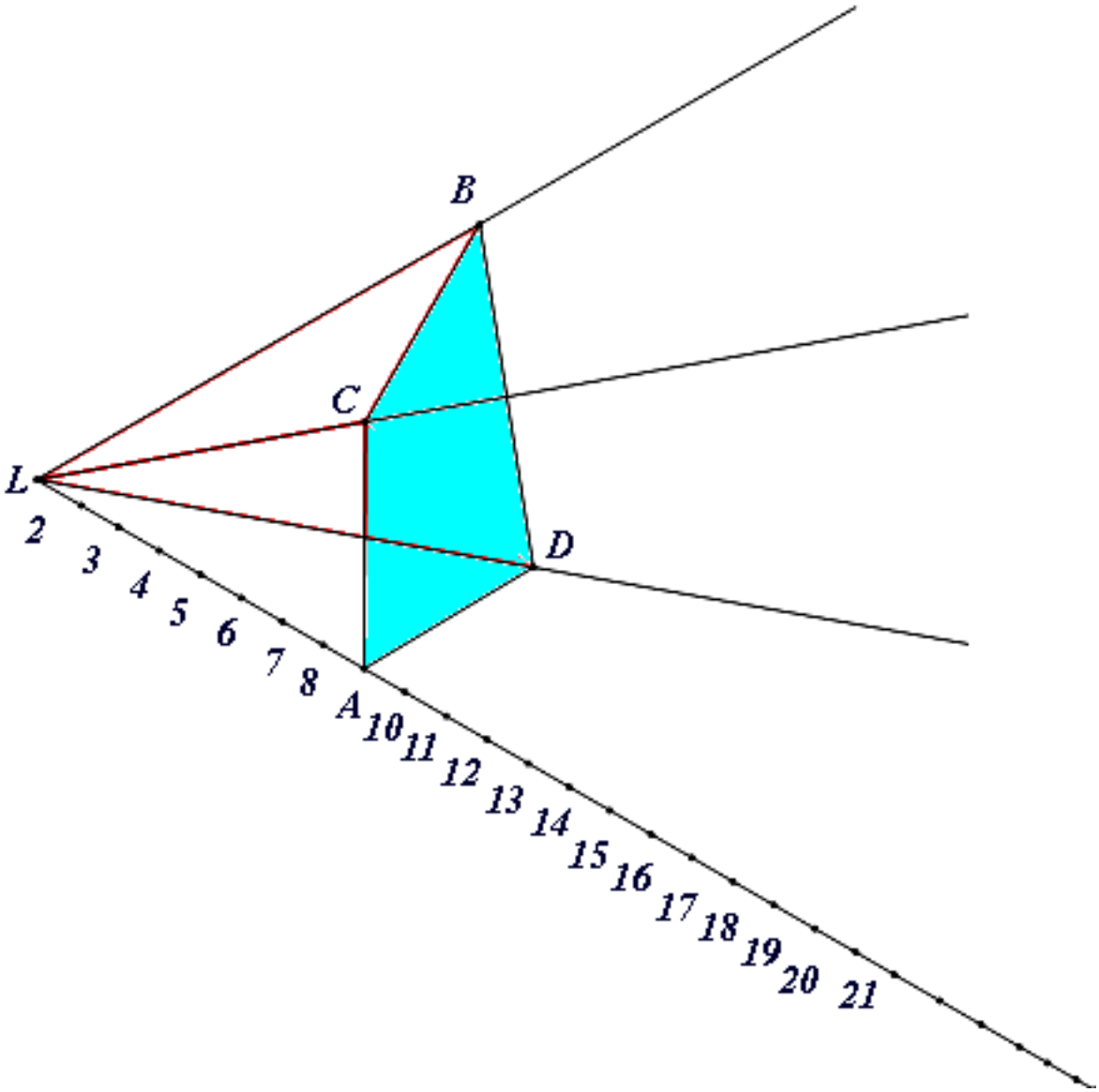


a Find the new length of AB.

b Find the new length of BC.

c Find the new length of CA.

16 Quadrilateral $ACBD$ is to be scaled by a certain factor. For each scale factor below, determine the new point that will correspond to point A.



a Scale factor of 3

b Scale factor of $\frac{1}{3}$

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17 A circular oil spill has a radius of $20\ \text{m}$. In a photo taken of the oil spill, the circle is reduced by a factor of 500.
Find the radius of the circular oil spill in the photo. Round to the nearest centimeter if necessary.

18 A rectangle is $8\ \text{cm}$ long and $6\ \text{cm}$ wide.
If its dimensions are enlarged such that its length is now $28\ \text{cm}$, find its new width.

19 An equilateral triangle of side length $6\ \text{cm}$ is to be enlarged by a factor of 5.

a Find the side length of the resulting triangle.

b Find the size of each angle in the resulting triangle.

20 Pythagorean triples such as $3, 4, 5$ represent lengths of possible right-angled triangles.
We want to make sure that a triangle with side lengths $15, 20$, and $25\ \text{cm}$ is also right-angled.

a Determine the enlargement factor needs to be applied to a triangle with side lengths measuring $3, 4$, and $5\ \text{cm}$, to get a triangle with side lengths measuring $15, 20$, and $25\ \text{cm}$.

b Show that a triangle with side lengths measuring $15, 20$, and $25\ \text{cm}$ is a right-angled triangle using Pythagoras' theorem.

21 William recorded a video on his camera. When viewing it on the camera screen, the video appeared in a width to length ratio of $4 : 3$ respectively. When he uploaded to his computer, everything appeared wider in the video.
Determine whether the following could be the ratio in which the video is shown on his computer.

a $20 : 12$ **b** $8 : 6$ **c** $16 : 12$ **d** $8 : 9$

e $10 : 7.5$

22 A small rectangle has a length of $7\ \text{cm}$ and an area of $21\ \text{cm}^2$. When it is enlarged, the resulting rectangle has a length of $21\ \text{cm}$.

a Determine the scale factor. **b** Find the area of the larger rectangle.

23 A square with area $4\ \text{cm}^2$ has its side lengths enlarged by a factor of 3.

a Find the side length of the original square.

b Find the area of the new square.

c Find the factor the area has been enlarged.

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24 The area of a square is $30\ \text{cm}^2$. Its side length is enlarged by a scale factor of 3.
Find the area of the new square.

25 A piece of sports tape in the shape of a rectangle measures $6\ \text{cm}$ in width and $10\ \text{cm}$ in length (when not stretched). When applied to Sophia's shoulder, it is stretched so that it covers a rectangular area measuring $12\ \text{cm}$ wide by $20\ \text{cm}$ long.

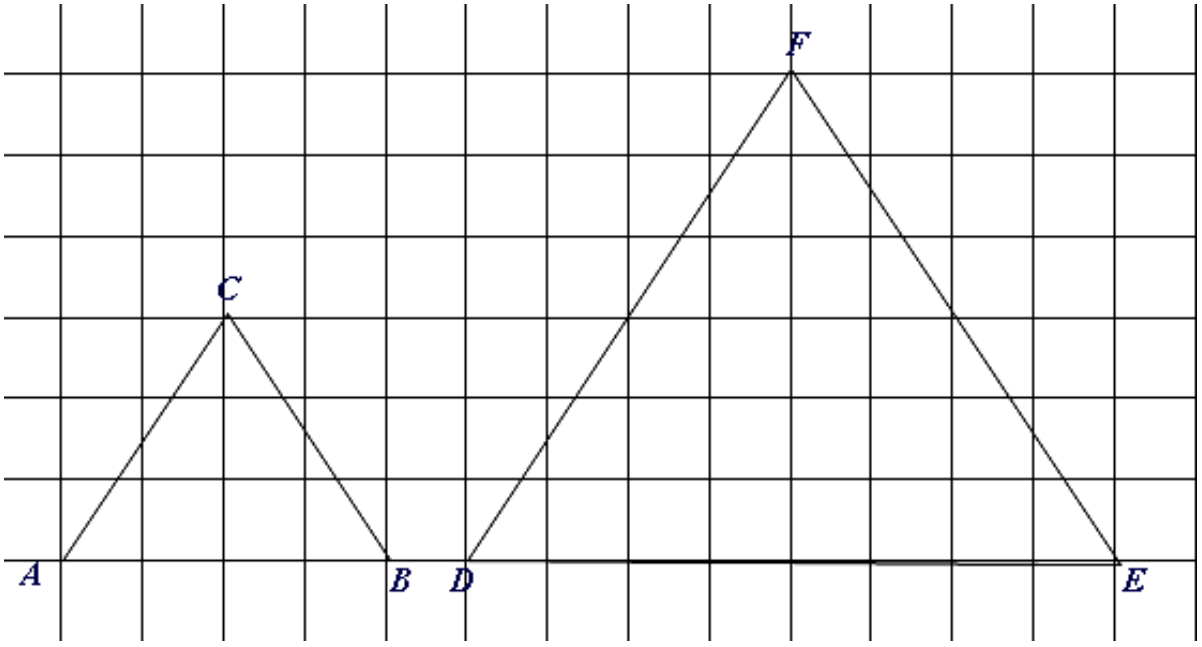
a Find the area the tape covers when it is not stretched.

b Find the area the tape covers when it is stretched.

c Find the scale factor the lengths of the rectangular tape is enlarged.

d Find the scale factor the area of the rectangular tape is enlarged.

26 Consider the two given triangles.



a Find the scale factor used to enlarge the small triangle.

b Find the scale factor used to reduce the large triangle.

c Find the area of the small triangle.

d Find the area of the large triangle.

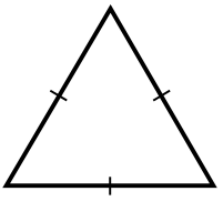
e Find the enlargement factor for the area of the small triangle.

27 Write the following using mathematical symbols:
$\triangle ABC$ is similar to $\triangle DEF$.

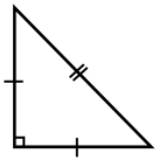
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28 Which pair of triangles are similar?

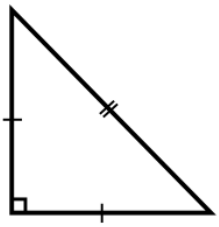
A



B



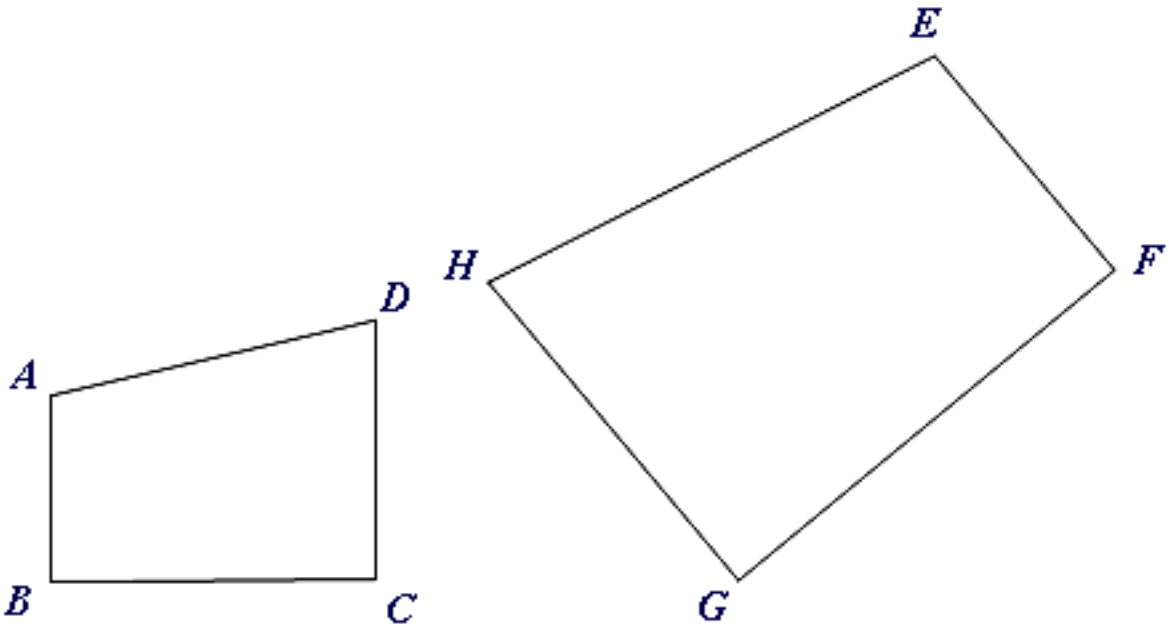
C



D



29 The smaller quadrilateral has been reflected, then enlarged and finally rotated.



a Determine whether two shapes are similar, congruent or niether.

b Which side in the larger shape is corresponding to $\overline{AB}$?

c Which angle in the larger shape is corresponding to $\angle ADC$?

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30 Dave wants to transform a shape but wishes it to remain similar to the original shape. Determine whether he can perform the following transformations while maintaining similarity.

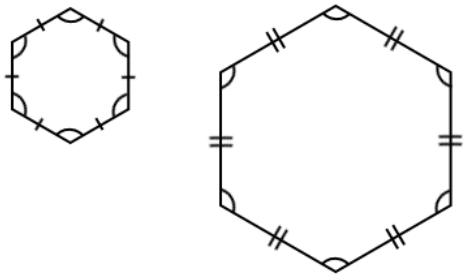
a Rotation

b Reflection

c Translation

d Enlargement

31 Consider the two shapes.



a Are the two shapes similar?

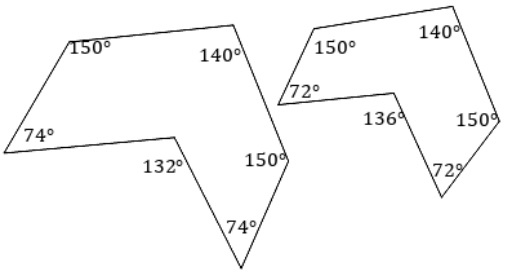
b State whether the following statements are true.

i One of the shapes is a rotation of the other.

ii One of the shapes is a translation of the other.

iii One of the shapes is an enlargement of the other.

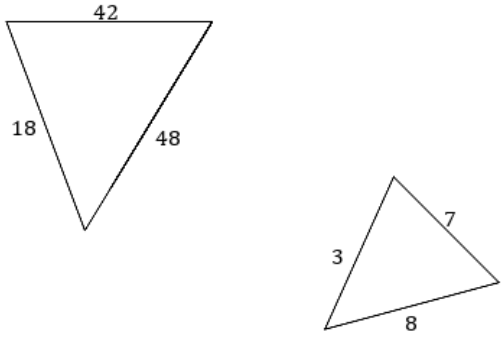
32 Determine whether the two shapes are similar.



33 Consider the shapes attached.

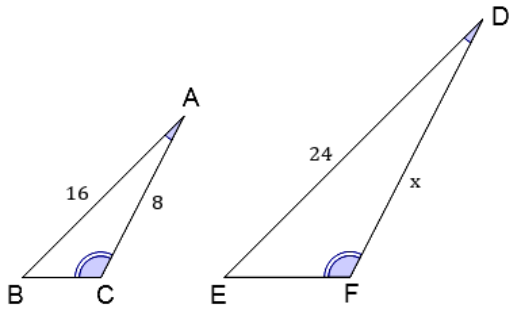
a Determine whether the two shapes are similar.

b Find the enlargement factor.



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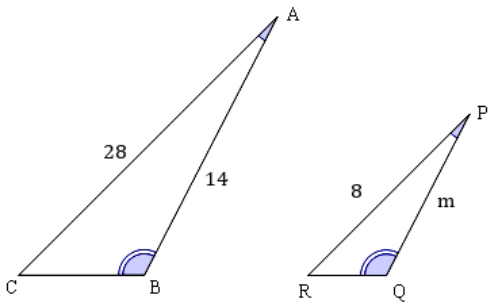
34 The two shapes in the following diagram are similar.



a Find the enlargement factor.

b Hence, find the value of x.

35 The two shapes in the following diagram are similar.



a Find the reduction factor.

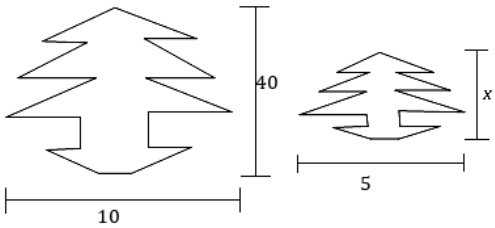
b Hence, find the value of m.

36 Given that each pair of following shapes are similar:

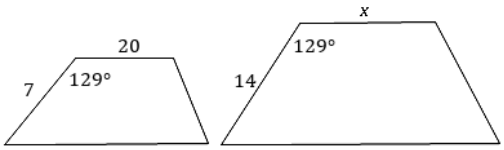
i Find the enlargement factor.

ii Find x.

a

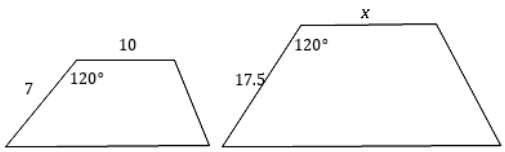


b

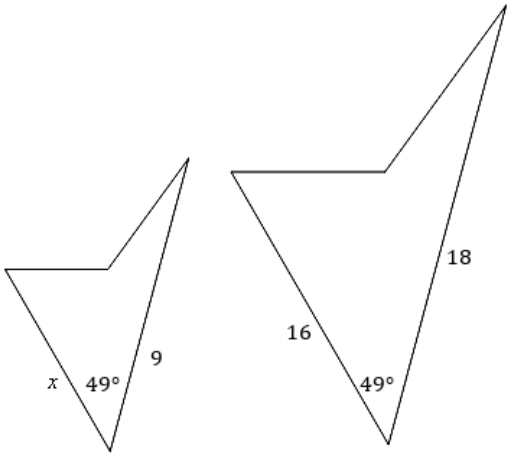


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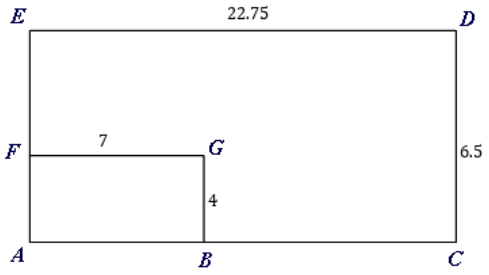
c



d



37 In the diagram, a smaller rectangle is inscribed within a larger rectangle with dimensions as given.
If $\overline{FG}$ remains fixed, find the length of $\overline{BG}$ so that the two rectangles are similar.



38 A square has a side length measuring 4 units and a rhombus has a side length measuring 8 units.

- **a** Are the sides in the same ratio?
- **b** Explain why the two shapes are not similar.

39 A square has side length $3\ \text{cm}$ and a rectangle has dimensions $18 \times 6\ \text{cm}$.

- **a** Are the corresponding angles equal?
- **b** Explain why the two shapes are not similar.

40 Determine whether the following statements are true.

- **a** If shape A is similar to shape B and shape B is similar to shape C, then shape A is similar to shape C.
- **b** Two congruent shapes are also similar.
- **c** If two shapes are similar, then they are also congruent.
- **d** All rectangles are similar.

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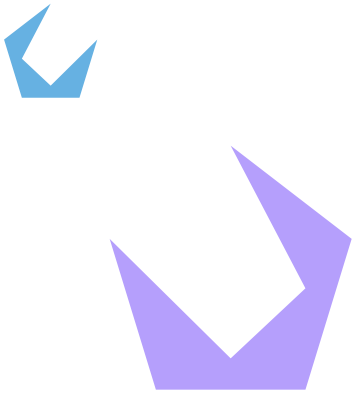
e All squares are similar.

f If every angle in an n-sided polygon matches with an angle in another n-sided polygon, then the polygons are definitely similar.

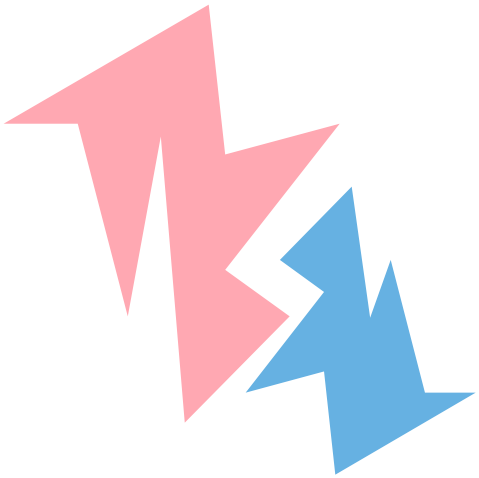
Additional questions

41 Determine whether each of the following pairs of shapes are similar.

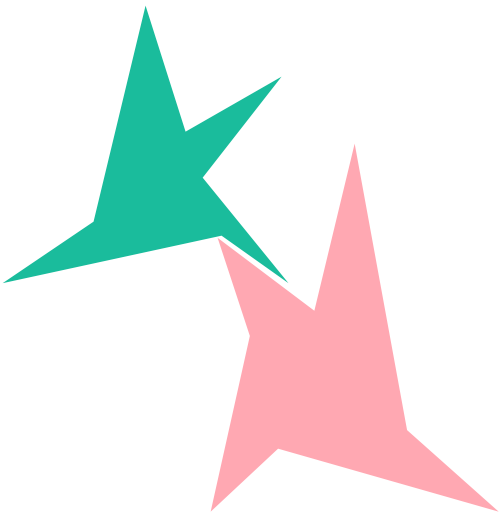
a



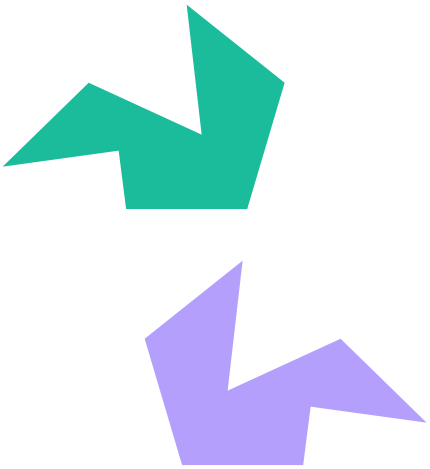
b



c

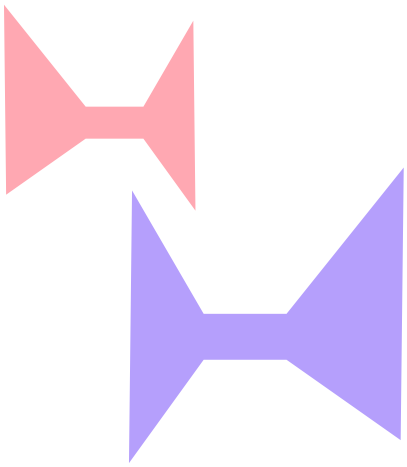


d



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e

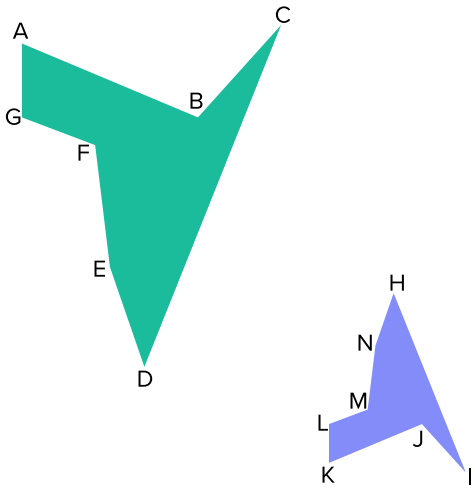


f

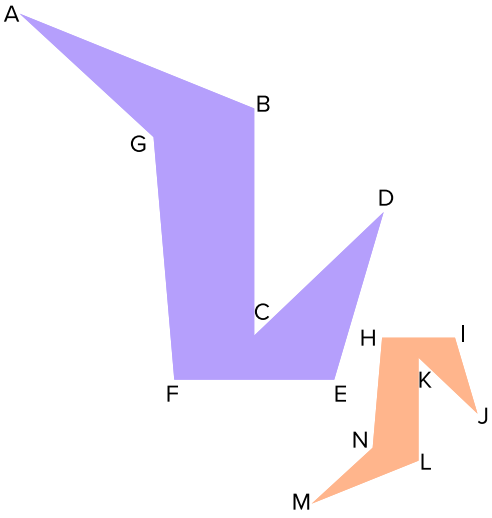


42 Each pair of following shapes are similar. For each pair identify the angle equal to $\angle CDE$.

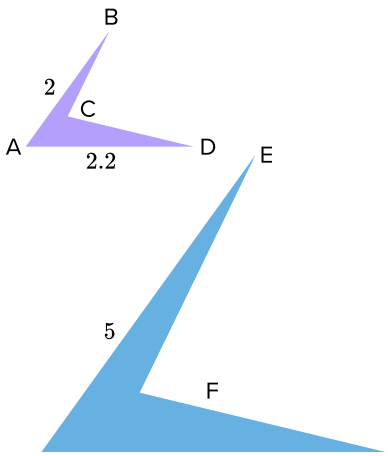
a



b



43 Quadrilaterals $ABCD$ and $EFGH$ are similar. Calculate GH.



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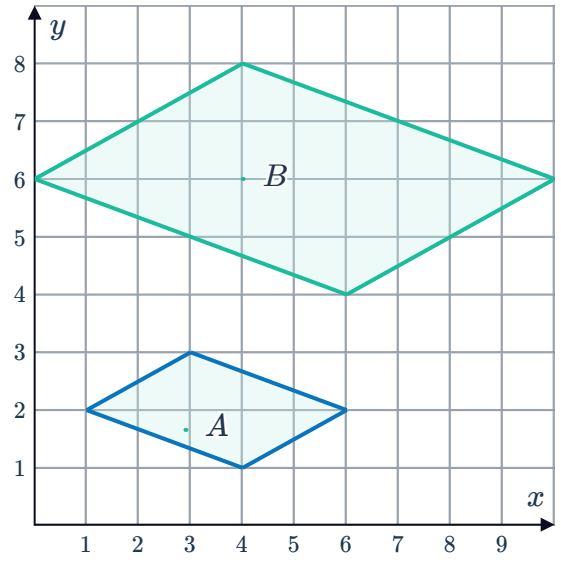
44 For each of the following pairs of shapes:

i Determine if A and B are similar.

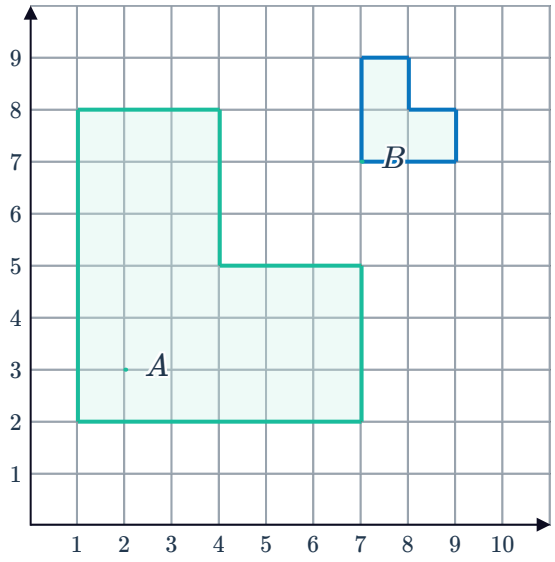
ii Determine if A and B are congruent.

iii If A and B are similar, identify the scale factor of the dilation transforming A to B.

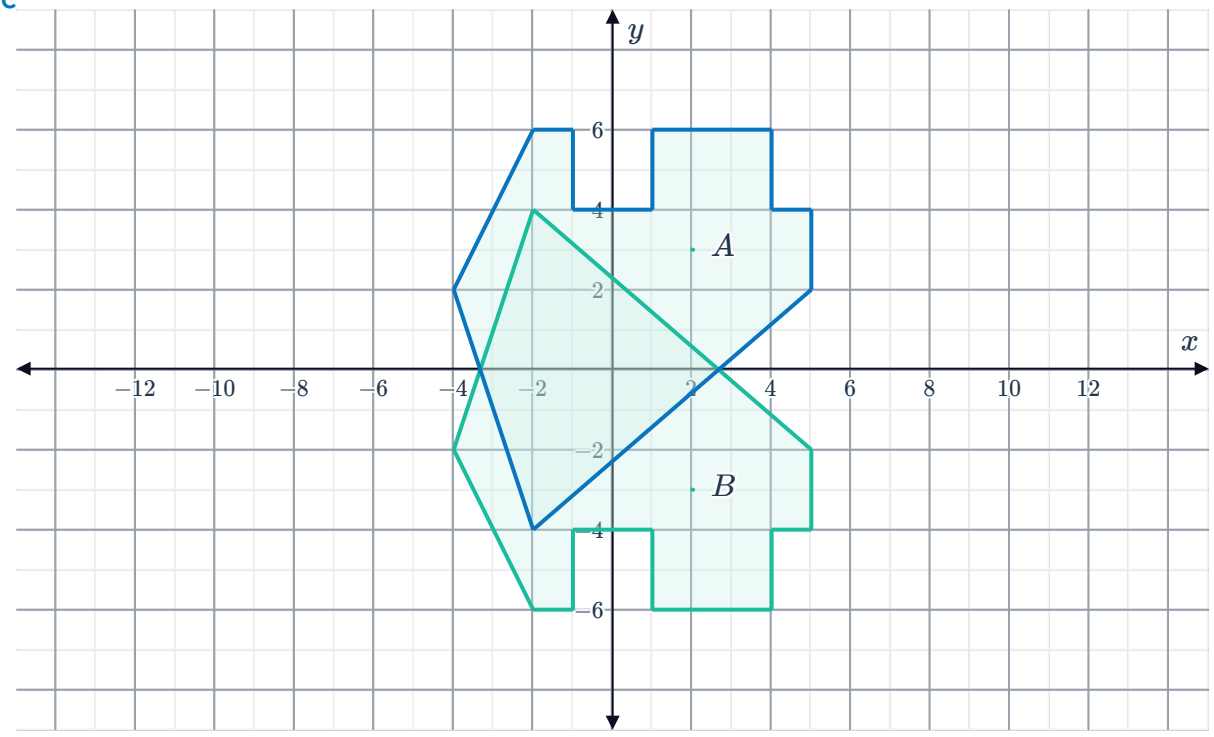
a



b



c



45 Which characteristics are required for two shapes to be similar?

46 What scale factor would be needed for two figures to be congruent after a dilation?

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